

conventions, and standards. To convert annual returns to weekly returns, $c = 1/52$. The expressions for annual returns can then be rewritten as expressions for weekly returns as follows:

$$R_{\text{weekly}} = (1 + R_{\text{daily}})^5 - 1; R_{\text{weekly}} = (1 + R_{\text{annual}})^{1/52} - 1. \quad (9)$$

One major limitation of annualizing returns is the implicit assumption that returns can be repeated precisely, that is, money can be reinvested repeatedly while earning a similar return. This type of return is not always possible. An investor may earn a return of 5 percent during a week because the market rose sharply that week, but it is highly unlikely that he will earn a return of 5 percent every week for the next 51 weeks, resulting in an annualized return of 1,164.3 percent ($= 1.05^{52} - 1$). Therefore, it is important to annualize short-term returns with this limitation in mind.

EXAMPLE 12

Annualized Returns

An analyst seeks to evaluate three securities she has held in her portfolio for different periods of time.

- Over the past 100 days, Security A has earned a return of 6.2 percent.
- Security B has earned 2 percent over the past four weeks.
- Security C has earned a return of 5 percent over the past three months.

1. Compare the relative performance of the three securities.

Solution:

To facilitate comparison, the three securities' returns need to be annualized:

- Annualized return for Security A: $R_{SA} = (1 + 0.062)^{365/100} - 1 = 0.2455 = 24.55\%$
- Annualized return for Security B: $R_{SB} = (1 + 0.02)^{52/4} - 1 = 0.2936 = 29.36\%$
- Annualized return for Security C: $R_{SC} = (1 + 0.05)^4 - 1 = 0.2155 = 21.55\%$

Security B generated the highest annualized return.

EXAMPLE 13

Exchange-Traded Fund Performance

An investor is evaluating the returns of three recently formed exchange-traded funds. Selected return information on the exchange-traded funds (ETFs) is presented in Exhibit 16.

Exhibit 16: ETF Performance Information

ETF	Time Since Inception	Return Since Inception (%)
1	146 days	4.61
2	5 weeks	1.10
3	15 months	14.35